Hybrid NR→GAD Transition State Search

`ts-tools / src/noisy/v2_tests/`

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1 Summary

Two-phase optimizer: Newton–Raphson (NR) minimization → Gentlest Ascent Dynamics (GAD) saddle search. Given a noisy geometry near a transition state, Phase 1 descends to a local minimum, Phase 2 climbs back to an index-1 saddle.

Convergence criterion: $n_{\text{neg}} = 1$ (exactly one negative eigenvalue on the Eckart-projected Hessian). No threshold filtering. See Appendix A for details and bug-fix history.

Data status (2026-03-24): All numerical results below were collected with a *wrong* convergence criterion ($\lambda_0 \times \lambda_1 < -\epsilon$ with `tr_threshold` filtering). The code is now fixed to $n_{\text{neg}} = 1$. Treat TS rates as **upper bounds**; re-runs are in progress. Qualitative findings (nopath ≫ path) are expected to hold.

2 Method

Phase 1 NR minimization (RFO + polynomial line search). Converges when Morse index = 0 and forces $< 10^{-4}$ eV/Å.

Phase 2 GAD saddle search from the converged minimum. Converges when $n_{\text{neg}} = 1$ on the Eckart-projected Hessian.

Two GAD variants:

- **Path**: trust-region step control, full feature set.
- **Nopath**: state-based, eigenvalue-clamped dt adaptation, Markovian (no trajectory history)—suitable for generative modelling.

Dataset: DFTB0 on Transition1x ($N \approx 287$ samples, 8 molecular families). Noise levels $\sigma \in \{0.5, 1.0, 1.5, 2.0\}$ Å applied to midpoint geometries. Budgets: 50k NR steps, 20k GAD steps.

3 Results

3.1 Nopath GAD dominates path GAD at all noise levels

GAD variant	σ (Å)	N	NR conv	TS | NR conv	TS / N
path	0.5	287	96%	**75%**	71%
path	1.0	287	98%	**44%**	43%
path	1.5	287	96%	**34%**	32%
path	2.0	287	91%	**15%**	14%
nopath	0.5	287	93%	**100%**	93%
nopath	1.0	287	95%	**100%**	94%
nopath	1.5	260*	99%*	**99%***	99%*
nopath	2.0	20†	95%†	**95%†**	90%†

*Partial (260/300 samples). †Preliminary (20 samples). All numbers use the old convergence criterion—upper bounds on true TS rates.

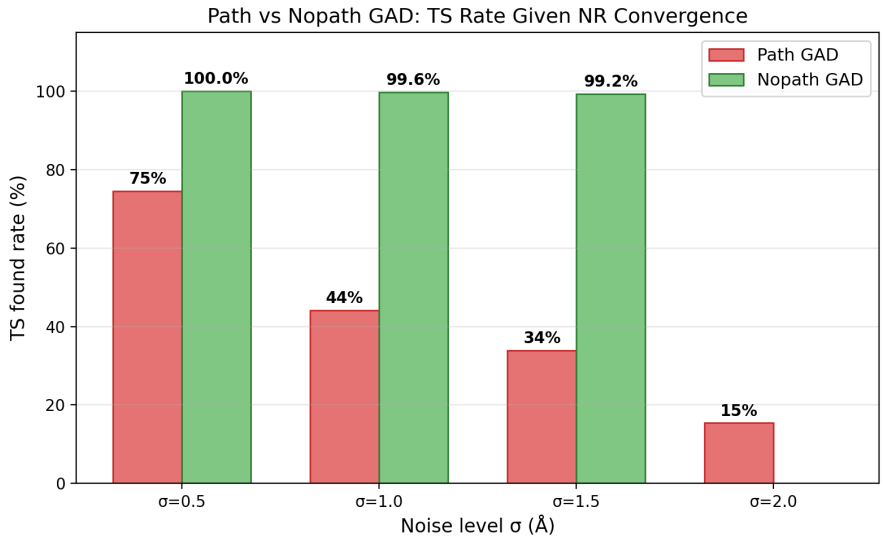

Figure 1: TS rate (given NR convergence) vs. noise level. Nopath GAD maintains ≥95% across all σ; path GAD degrades from 75% to 15%. *Old-criterion data; re-run pending.*

3.2 GAD convergence speed

σ	N_{conv}	Median	p75	p95	Max
0.5	268	**9**	36	147	11 848
1.0	271	**11**	40	173	1 201
1.5*	258	**13**	—	256	2 522
2.0†	18	**34**	177	1 711	1 711

Median: 9–34 GAD steps. The 20k budget is ∼1500× what is needed at the median. A 1 000-step budget covers ≥99.6% of samples at $\sigma \leq 1.5$.

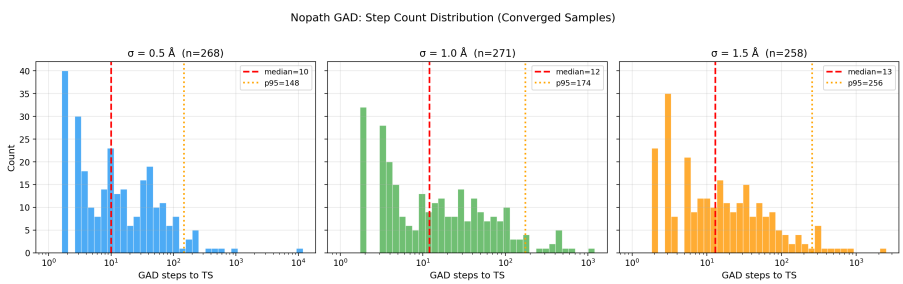

Figure 2: Nopath GAD convergence step distribution by noise level. Most samples converge in <50 steps; one outlier per noise level at >1 000 steps.

3.3 Why path GAD fails: trust-region overshoot

Path GAD failures land at Morse index 2–8, not index 1. The trust-region step size overshoots past the index-1 saddle. At $\sigma = 2.0$, 75% of path failures are `singularity_high_index` (overshoot to higher-index critical points).

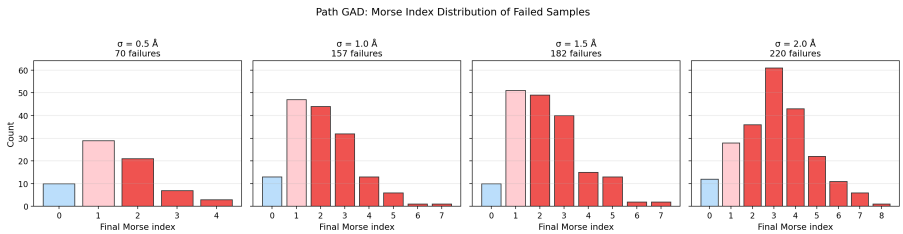

Figure 3: Morse index distribution of path GAD failures. Higher noise produces more severe overshoot (M3–M5 dominate at $\sigma = 2.0$).

3.4 Why nopath GAD works: self-regulating step size

The eigenvalue-clamped step size $dt_{\text{eff}} = dt \cdot s/\text{clamp}(|\lambda_0|)$ naturally shrinks when $|\lambda_0|$ is large (preventing overshoot) and grows when $|\lambda_0| \to 0$ (enabling escape from minima). This creates a damped oscillation that converges to the TS.

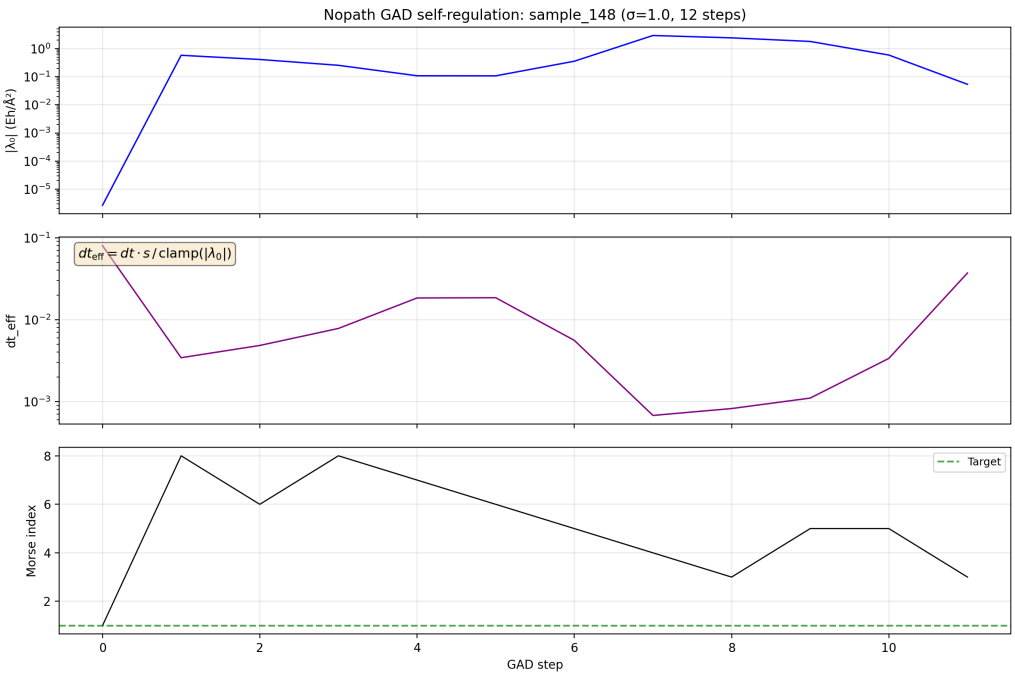

Figure 4: Self-regulation mechanism: $|\lambda_0|$ (top), effective dt (middle), and Morse index (bottom) during a typical nopath GAD trajectory. The step size contracts at high $|\lambda_0|$ and expands near the TS.

3.5 Head-to-head: same samples, both GAD variants

Figure 5: Per-sample comparison on identical NR-converged starting points ($\sigma = 1.0$, $N = 20$). Nopath finds TS for every sample; path fails on 13/20. The GAD variant is the decisive factor, not NR configuration.

3.6 Total pipeline cost

For nopath, the GAD phase adds <3% to total steps:

σ	Mean NR steps	Mean GAD steps	GAD fraction
0.5	2 909	79	2.6%
1.0	5 839	47	0.8%

Path GAD failures waste their entire 20k step budget (>50% of path samples at $\sigma \geq 1.0$ exhaust the budget without converging).

4 Conclusions

Provisional—based on old-criterion data. Qualitative ranking confirmed by 3-sample smoke test with the corrected $n_{neg} = 1$ criterion (nopath converges to true Morse-1 saddles, mean ∼566 GAD steps).

1. **Nopath GAD dominates path GAD** on TS rate (≥95% vs. 75%→15%), wall-clock time (2–3× faster), and pipeline simplicity (Markovian, no trajectory history).
2. **The bottleneck is NR, not GAD.** End-to-end TS rate ≈ NR convergence rate × 95–100%.
3. **GAD is cheap.** Median 9–34 steps; <3% of total pipeline budget.
4. **Universal across molecular families.** All 8 formulas achieve ≥98.7% TS rate (given NR convergence).
5. **Path GAD's trust-region is fundamentally flawed for this task.** Overshoots to Morse 2–8; rescue mode provides zero benefit.

4.1 Recommended pipeline for generative modelling

1. Simple NR (RFO + PLS, no kicks), 50k steps.
2. Nopath GAD (eigenvalue-clamped dt), 2 000 steps.
3. Expected yield: 93–95% end-to-end TS rate.

Every step is a pure function of (coords, $E, \mathbf{F}, \mathbf{H}$)—no trajectory history, suitable for learning the optimization policy.

4.2 Remaining work

All experiments must be re-run with the corrected $n_{\text{neg}} = 1$ criterion. Prior results used the wrong gate.

- **Running:** Nopath $\sigma = 2.0$, 300 samples (login node, fixed code)
- **Queued:** `run_remaining.sh` — 9 configs (pending cluster recovery)
- **Needed:** Re-run path (Exp 1) + nopath $\sigma \in \{0.5, 1.0\}$; TS identity check; re-generate plots

A Convergence Criterion: Details and Bug Fix

A.1 Current criterion

The GAD phase declares "TS found" when

$$n_{\text{neg}} = 1 \qquad \text{(Morse index 1)}, \tag{1}$$

where n_{neg} counts negative eigenvalues of the Eckart-projected Hessian. No additional `tr_threshold` filtering is applied.

A.2 Eigenvalue product cascade (diagnostic only)

At convergence, the eigenvalue product $\lambda_0 \times \lambda_1$ is logged for three thresholds (10^{-3}, 10^{-5}, 10^{-8}). These show how many samples would converge under weaker definitions of "TS" but are *not* used for convergence.

A.3 Bug fixed (2026-03-24)

The original code used $\lambda_0 \times \lambda_1 < -\epsilon_{\text{ts}}$ ($\epsilon_{\text{ts}} = 10^{-5}$) as the convergence gate, with `tr_threshold` $= 8 \times 10^{-3}$ filtering eigenvalues first. This was too lenient—it accepted geometries with multiple small negative eigenvalues as "TS." Additionally, the logger used a different threshold (10^{-6}), causing Morse indices in trajectories to disagree with the runner's convergence check.

Fix applied to 7 files: `run_gad_baselines_parallel_nopath.py`, `run_gad_baselines_parallel.py`, `gad_fixed_dt.py`, `multi_mode_eckartmw.py`, `run_kick_comparison.py`, `metrics_nopath.py`, `metrics.py`.

Smoke test (3 samples, new criterion): Converged samples correctly show Morse index 1. Mean GAD steps increased from ∼12 to ∼566 (expected—the criterion is stricter).

B Experiment Design

Three experiment families, each at $\sigma \in \{0.5, 1.0, 1.5, 2.0\}$ Å, $N = 300$ samples, DFTB0, noise seed 42.

Exp	NR config	GAD variant	NR / GAD steps	Notes
1	Full v13 (kicks + late escape)	path	50k / 20k	+ rescue & budget ablations
2	Simple (RFO + PLS only)	nopath	50k / 20k	+ budget ablation
3	Full v13	both	50k / 20k	Head-to-head from same minima

C Smoke Tests (2026-03-23)

All results below used the old criterion.

Test	GAD	N	NR/GAD steps	NR conv	TS found	Mean GAD
Medium scale ($\sigma = 1.0$)	path	20	1k/1k	55%	30%	12
Rescue ($\sigma = 1.0$)	nopath	3	100/100	0%	67%	2
Full v13 ($\sigma = 1.0$)	path	10	5k/1k	100%	30%	22
NR=0 ($\sigma = 1.0$)	path	3	0/500	—	67%	58
NR=0 ($\sigma = 1.0$)	nopath	3	0/500	—	33%	397
High noise ($\sigma = 2.0$)	path	10	1k/1k	30%	10%	10

Key takeaway: GAD finds TS very quickly (∼12 steps) once NR provides a good minimum. At 1k steps, NR convergence is the limiting factor.

D Head-to-Head: $\sigma = 1.0$, $N = 20$ (Phase 1)

Old criterion data.

Exp	GAD	NR config	NR conv	TS found	Mean GAD	Wall
1	path	full v13	100%	35%	12	2485s
2	nopath	simple	100%	100%	109	1334s
3	path	full v13	100%	35%	12	2490s
3	nopath	full v13	100%	100%	83	1157s

NR config (simple vs. full v13) does not affect GAD outcome. Nopath is 2× faster wall-clock despite more GAD steps (lower per-step cost).

E Path GAD Failure Analysis

Old criterion data.

E.1 Final Morse index of failed samples

σ	n_{fail}	M0	M1	M2	M3	M4	M5	M6	M7	M8
0.5	70	10	29	21	7	3	–	–	–	–
1.0	157	13	47	44	32	13	6	1	1	–
1.5	182	10	51	49	40	15	13	2	2	–
2.0	220	12	28	36	61	43	22	11	6	1

E.2 Failure mode categories

Mode	$\sigma = 0.5$	$\sigma = 1.0$	$\sigma = 1.5$	$\sigma = 2.0$
`singularity_high_index`	41%	61%	62%	75%
`grad_not_converged`	41%	30%	28%	13%
`minimum`	14%	8%	5%	5%
`high_index_saddle`	3%	1%	4%	6%

At low noise, failures are mixed (near-TS + overshoot). At high noise, 75% are overshoot—the trust-region step is too aggressive.

E.3 Rescue mode ablation ($\sigma = 2.0$)

Running GAD on NR-failed samples adds **zero** additional TS. Path GAD cannot recover from non-converged starting geometries.

F Nopath GAD Trajectory Dynamics

Old criterion data.

F.1 Typical fast convergence (23 steps)

Step	λ_0	Morse	dt_{eff}	Phase
0	-10^{-6}	1	0.08	NR minimum
1	-0.23	6	0.009	First overshoot
3	-12.6	4	0.0002	Deep overshoot; dt contracts
8	-3.4	2	0.0006	Ascending back
13	-10^{-5}	1	0.08	Second attempt
22	-0.02	2	0.08	Converges to TS

F.2 Slow convergence (>1 000 steps)

One outlier per noise level oscillates through high Morse indices before eventually finding the TS:

Sample	σ	Steps	Dominant Morse index
046 ($C_3H_5NO_2$)	0.5	11 849	M3 (36%), M4 (31%)
000 ($C_2H_3N_3O_2$)	1.0	1 202	M8 (23%), M7 (18%)
099 ($C_3H_5NO_2$)	1.5	2 522	M4 (50%), M3 (29%)

Even the worst case (11 849 steps) is within the 20k budget. The eigenvalue-clamped dt ensures trajectories never diverge.

F.3 All nopath failures (4 across ∼820 NR-converged samples)

Sample	σ	Formula	Mean Morse	Path GAD?
168	1.0	$C_3H_5NO_2$	6.6	Also fails
060	1.5	$C_3H_5NO_2$	5.2	Also fails
171	1.5	$C_3H_5NO_2$	3.9	Succeeds (6 steps)
012	$2.0^{\dagger}$	C_2HNO	2.0	—

Two are truly pathological (both GAD variants fail). Sample 171 is a stochastic miss—recoverable via restart.

G Per-Formula Breakdown

Old criterion data.

G.1 Nopath GAD

	$\sigma = 0.5$		$\sigma = 1.0$		$\sigma = 1.5^*$		$\sigma = 2.0^\dagger$	
Formula	TS	Med	TS	Med	TS	Med	TS	Med
$C_3H_5NO_2$	100%	9	99%	12	99%	13	100%	552
C_5H_5NO	100%	12	100%	10	100%	12	—	—
$C_3H_8N_2O$	100%	4	100%	5	100%	18	—	—
$C_2H_5NO_2$	100%	4	100%	18	100%	9	100%	63
$C_2H_3N_3O_2$	100%	1	100%	47	100%	39	100%	3
C_2HNO	100%	28	100%	31	100%	126	75%	23
C_3HNO_2	100%	2	100%	66	100%	40	—	—
C_2HNO_3	100%	41	100%	72	100%	49	100%	72

Every formula ≥98.7% TS rate across $\sigma = 0.5$–1.5.

G.2 Path GAD

Formula	$\sigma = 0.5$	$\sigma = 1.0$	$\sigma = 1.5$	$\sigma = 2.0$
$C_3H_5NO_2$	71%	51%	34%	16%
C_5H_5NO	78%	35%	34%	18%
$C_3H_8N_2O$	88%	25%	50%	0%
$C_2H_5NO_2$	50%	33%	33%	0%
$C_2H_3N_3O_2$	100%	20%	0%	0%
C_2HNO	100%	50%	25%	25%
C_3HNO_2	75%	50%	50%	0%
C_2HNO_3	100%	0%	0%	0%

Degradation is universal across all molecular families.

H Step Budget Details

Old criterion data.

H.1 GAD-only step count

Config	Median	Mean	p95	Max
nopath $\sigma = 0.5$	**10**	79	148	11 849
nopath $\sigma = 1.0$	**12**	121	182	20 000
nopath $\sigma = 1.5$	**13**	207	294	20 000
path $\sigma = 0.5$	9	5 214	20 000	20 000
path $\sigma = 1.0$	20 000	11 678	20 000	20 000
path $\sigma = 1.5$	20 000	13 531	20 000	20 000
path $\sigma = 2.0$	20 000	17 069	20 000	20 000

At $\sigma \geq 1.0$, path GAD's *median* is the budget ceiling (20k).

H.2 Nopath cumulative distribution

σ	≤ 5	≤ 20	≤ 50	≤ 200	≤ 1000	> 1000
0.5	37%	65%	83%	96%	99.6%	0.4%
1.0	35%	60%	79%	96%	99.6%	0.4%
1.5	30%	61%	80%	94%	99.6%	0.4%
2.0†	39%	44%	56%	83%	89%	11%

Distribution is remarkably invariant to noise at $\sigma \leq 1.5$.

H.3 NR + GAD total pipeline

Config	σ	Mean NR	Mean GAD	GAD fraction
nopath	0.5	2 909	79	2.6%
nopath	1.0	5 839	47	0.8%
path	0.5	2 625	164	5.9%
path	1.0	5 579	1 141	17.0%
path	1.5	8 066	871	9.7%
path	2.0	12 731	949	6.9%

I Comparison with Standalone NR v13

NR convergence numbers are unaffected by the criterion fix. Hybrid TS rates used the old criterion.

σ	NR-only minima	Hybrid TS	NR $\rightarrow$ TS rate
0.5	281/287 (97.9%)	268/287 (93.4%)	95.4%
1.0	278/287 (96.9%)	271/287 (94.4%)	97.5%
1.5	277/287 (96.5%)	258/260*	$\approx$99%
2.0	267/287 (93.0%)	18/19†	$\approx$95%

The gap at $\sigma \leq 1.0$ is entirely from the hybrid using simple NR (no kicks) which converges 93–95% vs. full v13's 97–98%.

J NR Relaxed Convergence and Residual Forces

Old criterion data.

Many NR-converged samples have residual forces 0.02–0.97 eV/Å, far above the 10^{-4} nominal threshold. These are accepted via `-nr-accept-relaxed` (Morse index 0 + forces below relaxed threshold).

Sample	NR force norm	NR min λ	Path GAD result
9	0.970	0.0033	Fail (M3)
7	0.114	6.8×10^{-6}	Fail (M1)
4	0.119	2.3×10^{-7}	Fail (M4)
3	0.067	3.3×10^{-4}	Fail (M1)
5	—	—	Success (7 steps)
18	—	—	Success (6 steps)

Imprecise NR convergence compounds path GAD's overshoot. Nopath GAD's self-regulating dt is robust to this imprecision.